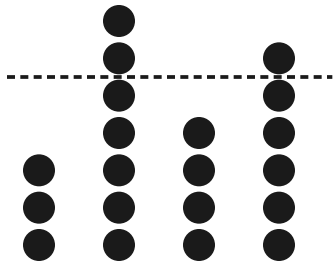
Mean, Median and Mode

Three ways to say what is typical.

Mean: even it out (a fair share)



Move dots from the tall towers to the short ones.
Now every tower has 5.

20 ÷ 4 = 5

Median: the middle one, in order

9, 2, 7, 4, 12 → 2, 4, 7, 9, 12

7 is in the middle. Two in the middle? Take the number halfway between them.

Mode: the one that comes up most. Range: biggest – smallest.

3, 5, 5, 8, 9: mode 5, range 9 − 3 = 6

Your turn!

mean of 4, 8, 9 = 7

For teachers

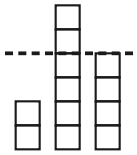
Goal: mean as a fair share, median, mode and range (6.SP.3, 6.SP.5c). Say “If we shared them out evenly, how many each?” For the median, always put the numbers in order first.

Mean, Median and Mode: Practice

Even it out, find the middle, find the most.

Part 1 Even it out Move cubes until the towers match.

1)

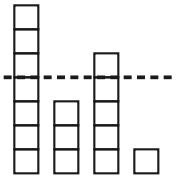


in all: 12

12 ÷ 3 = 4

mean = 4

2)

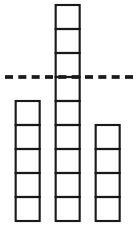


in all: 16

16 ÷ 4 = 4

mean = 4

3)

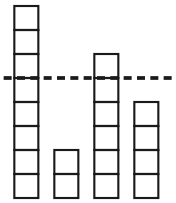


in all: 18

18 ÷ 3 = 6

mean = 6

4)

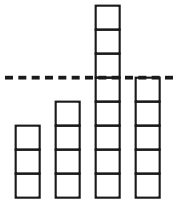


in all: 20

20 ÷ 4 = 5

mean = 5

5)

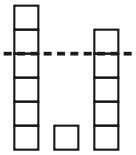


in all: 20

20 ÷ 4 = 5

mean = 5

6)



in all: 12

12 ÷ 3 = 4

mean = 4

Part 2 Find the median Write them in order. Ring the middle one.

7) 8, 3, 11, 6, 5

3, 5, **(6)**, 8, 11 median = 6

8) 14, 9, 12, 20, 7

7, 9, **(12)**, 14, 20 median = 12

9) 4, 10, 2, 8, 6, 3, 9

2, 3, 4, **(6)**, 8, 9, 10 median = 6

Mean, Median and Mode: Practice (continued)

Part 2 Find the median (continued)

10) **25, 18, 31, 22, 27**

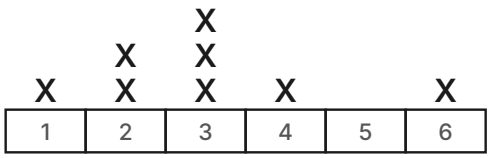
18 | 22 | 25 | 27 | 31 **median =** 25

11) **6, 13, 9, 2, 11**

2 | 6 | 9 | 11 | 13 **median =** 9

Part 3 Read the X chart Find the middle, the most, and the spread.

12) **Goals scored**



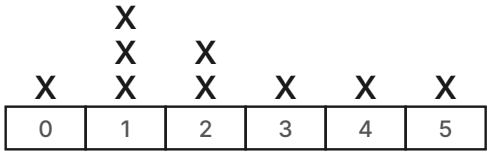
Median? 3

Mode? 3

Range? 5

Mean? 3

13) **Pets at home**



Median? 2

Mode? 1

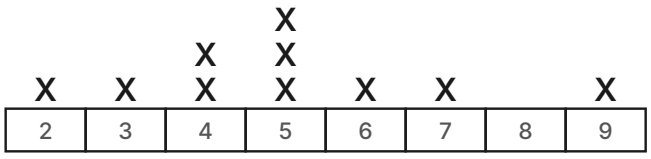
Range? 5

How many? 9

Mean, Median and Mode: Practice (continued)

Part 3 Read the X chart (continued)

14) **Books read**



Median? **5**

Mode? **5**

Range? **7**

Mean? **5**

Part 4 Mean Add them, then share them out.

15) **mean of 5, 9, 10 = 8**

16) **mean of 12, 4, 8 = 8**

17) **mean of 3, 7, 6, 8 = 6**

18) **mean of 15, 9, 21 = 15**

19) **mean of 6, 6, 11, 9 = 8**

20) **mean of 20, 14, 17 = 17**

21) **mean of 2, 9, 5, 12 = 7**

22) **mean of 30, 18, 24 = 24**

Mean, Median and Mode: Practice (continued)

Part 5 Median In order first.

23) median of 7, 2, 9 = 7

24) median of 15, 4, 11, 8, 6 = 8

25) median of 3, 12, 5 = 5

26) median of 21, 16, 19, 25, 30 = 21

27) median of 8, 4, 10, 6 = 7

28) median of 13, 7, 9, 11 = 10

Part 6 Mode and range The most often, and biggest − smallest.

29) mode of 4, 7, 7, 2, 9 = 7

30) range of 13, 5, 22, 9 = 17

31) mode of 11, 3, 8, 3, 6 = 3

32) range of 40, 18, 27, 33 = 22

33) mode of 9, 12, 12, 15, 12 = 12

Mean, Median and Mode: Practice (continued)

Part 6 Mode and range (continued)

34) range of 6, 19, 2, 11 = **17**

Part 7 The missing number What makes the mean right?

35) mean of 6, 8, **13** = 9

36) mean of 10, 4, **10** = 8

37) mean of 7, 9, 5, **11** = 8

38) mean of 12, 15, **6** = 11

Part 8 More means

39) mean of 8, 14, 2 = **8**

40) mean of 23, 27, 16 = **22**

41) mean of 9, 4, 29, 6 = **12**

42) mean of 6, 9, 9 = **8**

43) mean of 30, 8, 18, 26, 28 = **22**

Mean, Median and Mode: Practice (continued)

Part 8 More means (continued)

44) **mean of 17, 26, 6, 19 = 17**

45) **mean of 10, 4, 26, 29, 11 = 16**

46) **mean of 28, 9, 29 = 22**

47) **mean of 27, 25, 19, 5 = 19**

48) **mean of 13, 2, 12 = 9**

49) **mean of 23, 22, 22, 13 = 20**

50) **mean of 29, 12, 2, 22, 20 = 17**

51) **mean of 27, 16, 20, 5 = 17**

52) **mean of 4, 10, 27, 11 = 13**

Part 9 More medians

53) **median of 14, 20, 2, 22, 9, 25, 29 = 20**

54) **median of 19, 4, 10, 31, 9 = 10**

Mean, Median and Mode: Practice (continued)

Part 9 More medians (continued)

55) median of 4, 21, 2, 15, 11 = 11

56) median of 19, 39, 14, 36 = 27.5

57) median of 34, 3, 29, 36 = 31.5

58) median of 12, 23, 18, 16, 29 = 18

59) median of 20, 35, 38, 23, 4 = 23

60) median of 29, 35, 9, 32, 13 = 29

61) median of 27, 34, 4, 30, 14 = 27

62) median of 13, 14, 35, 21, 39, 18, 6 = 18

63) median of 20, 23, 4, 6, 10, 12, 7 = 10

64) median of 3, 29, 36, 24 = 26.5

Mean, Median and Mode: Practice (continued)

Part 10 More modes and ranges

65) **range of 21, 32, 4, 11, 44 = 40**

66) **range of 32, 31, 41, 21, 5 = 36**

67) **range of 21, 9, 11, 22, 48 = 39**

68) **range of 20, 47, 11, 27, 13 = 36**

69) **range of 18, 3, 9, 32, 44 = 41**

70) **range of 5, 14, 43, 44, 16 = 39**

71) **range of 6, 13, 10, 33, 7 = 27**

72) **range of 9, 17, 21, 41, 32 = 32**

73) **range of 48, 25, 19, 15, 16 = 33**

74) **range of 15, 10, 23, 14, 39 = 29**

Mean, Median and Mode: Practice (continued)

Part 11 Remember? Problems from earlier packets.

75) $14 \times 9 \div 2 =$ **63**

76) $\frac{1}{2} \times 6 \times 9 =$ **27**

77) $9x = 117$: $x =$ **13**

78) 45% of 60 = **27**

79) $5.4 \times 0.3 =$ **1.62**

80) $\frac{2}{3} \div \frac{1}{6} =$ **4**

Part 12 Word problems Mean, median, mode or range?

81) Ana scored 8, 6 and 10 points in three games. What is her mean score?

Number sentence **8 + 6 + 10 = 24, 24 ÷ 3 = 8**

Answer **8** points

Draw or show your work

82) Five plants are 12, 9, 15, 10 and 14 cm tall. What is the median height?

Number sentence **9, 10, 12, 14, 15**

Answer **12** cm

Draw or show your work

83) Shoe sizes in a group: 4, 5, 5, 6, 5, 7. What is the mode?

Number sentence **5 comes up 3 times**

Answer **5**

Draw or show your work

Mean, Median and Mode: Practice (continued)

Part 12 Word problems (continued)

84) The coldest day was 18 degrees and the warmest 41. What is the range?

Number sentence **41 − 18 = 23**

Answer **23** degrees

Draw or show your work

85) Challenge: Leo's mean on 3 tests is 80. He got 75 and 85 on two. What was the third?

Number sentence **240 − 160 = 80**

Answer **80** points

Draw or show your work

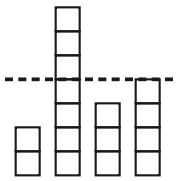
Check-In: Mean, Median and Mode

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Even it out

1)

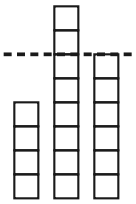


in all: 16

16 ÷ 4 = 4

mean = 4

2)



in all: 18

18 ÷ 3 = 6

mean = 6

Part 2 Mean

3) **mean of 4, 9, 11 = 8**

4) **mean of 16, 10, 22 = 16**

Part 3 Median

5) **median of 9, 3, 14, 6, 11 = 9**

6) **median of 18, 12, 25 = 18**

Part 4 Mode and range

7) **range of 17, 8, 25, 12 = 17**

8) **mode of 6, 9, 9, 4, 12 = 9**